Final project topic

Due date: January 11th 2026

1. Band structure calculations of two-dimensional Kagome lattice by tight binding model
2. Tight band model in a 1D chain
3. Density-function theory for band structure calculations
4. Pseudopotentials method for band structure calculations
5. The $k \cdot p$ theory for band structure calculations
6. Hubbard model and Mott insulators
7. Z2 topological insulators
8. Landau Fermi liquid theory
9. Su-Schrieffer-Heeger model
10. Kitaev chain and Topological 1D superconductor
11. Berry phase and Berry curvatures
12. Boltzman theory of electron transport
13. Anderson localization of charge carriers
14. Haldane model of graphene
15. Kane-Mele model of graphene
16. Bistritzer-MacDonald model of magic-angle twisted bilayer graphene
17. Bloch electron under magnetic field and Hofstadter's butterfly
18. Electronic and mechanical properties of carbon nanotubes
19. Transport in quantum wires and quantum dots

Final exam

1. Close books exam. But everyone can bring one paper of notes with a A4 size

2. All the questions and problems are from or extended from lecture slides or homework

3. The examination mainly examines the basic concepts.

时间：2025 年 12 月 30 日星期二 8:50-10:50

地点：E10-205 云谷校区

监考人：徐水钢、项寒笑